Irregular Labor Supply and Coordination in Firms: Experimental Evidence from India

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Abstract

Firms in developing countries face irregular labor supply – uncertainty in whether and when workers will arrive at work. Such uncertainty is especially costly when production requires coordinating labor inputs. This paper uses a randomized experiment with 92 small manufacturing firms in India to examine how irregular labor supply affects firms' ability to coordinate production. In the status quo, three in five workers arrive after their firm's start time on average. I incentivize workers at 20% of their daily earnings to arrive by the start time for four weeks. Incentives increase timely arrival from 41% to 59% of worker-days, driven by these late workers. Worker output increases by 24% for all workers in treatment firms, not only those who changed their arrival times as a result of the incentive. I show that these spillovers come from gains along the production line due to improved coordination. Treatment firms produce 29% more output, receive 21% more new orders and are 40% more likely to complete orders on time.

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1 Introduction

Firm growth is characterized by a shift from small, self-employment units into larger, more complex organizational forms that leverage complementarities between production inputs (Kremer, 1993; Gollin, 2008). Seminal work documenting that firms in low- and middle-income countries remain predominantly small and less productive relative to their counterparts in richer countries has motivated a large and growing literature studying barriers to firm growth in these settings (Hsieh and Klenow, 2009; Quinn and Woodruff, 2019). However, there is little research documenting whether and how these barriers affect the transition towards more complex organizational forms. This gap stems largely from a lack of data on the internal organization of firms, owing to limited record keeping and measurement challenges, with a notable exception of Bassi et al. (2023).

This paper highlights a novel labor market friction – worker tardiness – as a barrier to firms' ability to effectively coordinate production, and its impact on firm outcomes. I use a randomized experiment to study this in the context of a large apparel manufacturing cluster in India comprising small-medium firms. Two features of this setting make it ideal for this study. First, in the status quo, workers are paid individual piece rates, and choose their own work schedules. As a result, worker arrival times at the firm are highly dispersed, ranging from 3 hours early to 5 hours late, with a 90-10 gap of more than 2 hours (see Figure 1). Second, firms are organized as production lines where each worker performs a distinct operation on the garment in batches of 80-100 pieces, which are passed sequentially from one worker to the next. This creates complementarities in the production process, so that the output of workers performing downstream operations depends on those performing upstream operations. The sequential nature of production and observed dispersion in worker arrival times raise the question of whether workers and the firm can gain from coordinating worker arrival times.

I conduct an experiment with 92 firms comprising 600 workers. I incentivize workers in treatment firms to follow fixed work start times for a period of 4 weeks. Treatment incentives are designed as a daily bonus of INR 100-150 (20% of average daily earnings), paid conditional on arriving at the firm by the owners' work start time (hereafter, preferred start time), elicited at baseline. Workers in control firms receive unconditional transfers during the treatment incentive period, matched to a randomly chosen treatment firm each week to address concerns about income effects on worker output.

I collect daily data on worker attendance, arrival times, task assignments and production output starting 2 weeks before the incentives, and the 4-week incentive period. This allows

me to estimate the effect of the incentives on labor supply (on-time arrival and daily work hours) and production outcomes (output and earnings) for both workers and firms using a difference-in-differences specification with worker and time fixed effects.

I find that the incentives successfully encourage workers to arrive on time in treatment firms. Timely arrival increases by 18 percentage points, a 43% increase relative to a baseline control mean of 41%. Given the average size of firms, this translates to 1 additional worker being present at the firm before the preferred start time. This effect does not reflect higher attendance rates, suggesting workers responded through the intensive margin of changing their arrival times. To identify who responded most to treatment, I classify workers as early, late, and very late based on their average arrival times during baseline weeks. The timely arrival effect is driven entirely by late and very late workers, on whom the incentive was most binding – late (very late) workers are 21 (35) percentage points more likely to arrive on time, a 54% (88%) increase relative to the average control worker. I find no evidence that early workers altered their labor supply in response to the incentives. Finally, the increase in work hours is small and not statistically significant. Workers in treatment firms work on average 16 minutes longer per day than control firm workers.

Overall, the increase in timely arrival in treatment firms results in a meaningful decline in dispersion in arrival times. Total dispersion in worker arrival times, measured as the difference in arrival times between the earliest and the last worker, decreases by 22 minutes (17% of the control mean). This is driven by the last worker arriving on average 14 minutes earlier in treatment firms, a 29% reduction relative to a control mean of 47 minutes. Incentives thus shift the distribution of worker arrival times at the firm to the left, while also improving coordination in worker arrival times.

As a result, workers in treatment firms produce 0.27 standard deviations more in weekly output than workers in control firms (24% increase relative to the control mean, equivalent to 900 more pieces per week).[1] Crucially, the incentives generated positive spillovers among workers in treatment firms: the increase in worker output is realized by all workers, including both early arrivals who did not respond to incentives, and those who arrived late at baseline and responded most to the incentives.

Next, I present evidence that these spillovers reflect gains from improved coordination along the production line. To do so, I classify firms as "Bottleneck" firms if any worker whose modal baseline task is upstream in the production line – an upstream task worker – is a late worker at baseline. Remaining firms are classified as "Non-Bottleneck" firms, where

[1] Worker output is standardized to pre-incentive, product-level means.

no upstream task worker arrives late at baseline. If there are gains from coordination, I expect to see output gains among early workers to be driven by Bottleneck firms. The effect on weekly output for the average worker is larger in Bottleneck firms (0.36 SD) than in Non-Bottleneck firms (0.12 SD, not statistically significant). More importantly, *which* workers experience this gain differs across the two samples. I find that spillovers from late to early workers only exist in Bottleneck firms – all workers experience a 0.39 SD increase in weekly output. Adding further support for the coordination channel, I find that in Non-Bottleneck firms, gains in worker output are concentrated only among late workers, with no increase in output for early workers.

Finally, gains in worker output may reflect either an increase in worker productivity, or a mechanical effect from working additional hours. I find that these gains do reflect an increase in productivity: output per hour increases by 5% on average for all workers, but the increase is largest among late workers, whose productivity increases by 9% (p-value = 0.09). I provide suggestive evidence that this increase stems from more productive time utilization among late workers. Late workers in treatment firms are 5 percentage points less likely to perform non-production tasks (a 33% reduction relative to control workers), and instead spend more time during the work day on tailoring tasks.

These results suggest that late arrivals at the firm hurt workers' own output, and generate a negative externality on early workers by creating hold-ups on the production line. This leads to two questions: why workers arrive late, and what production line hold-ups imply at the firm level. Both workers and owners cite a preference for flexible work schedules as the main reason for late arrivals. Survey responses suggest that this preference reflects a combination of gender norms around household work (which are more binding for women), poor infrastructure reliability, and participation in local cultural/social events. Late workers are more likely to be women, and thus, the labor supply response for women is twice as large as that for men.[2,3] Consistent with results for early versus late workers, I find that men and women in treatment firms gain equally in weekly output despite responding differentially in labor supply, suggesting that female workers had been imposing a negative externality on male workers in the firm.

[2]The preference for flexibility extends beyond women. Late workers are also more likely to be married, and have young children in the household. Consequently, late workers who are male commonly cite the need to drop off or pick up their children to/from school as the reason for lateness, owing to the lack of, or limited reliability of school provided transportation.

[3]This is consistent with prior work documenting that workers in developing countries have a greater demand for flexibility, given the prevalence of health shocks (Breza and Kaur, 2025), social pressure to attend cultural/family events (Goraya et al., 2025), and gender norms around domestic responsibilities (Ho et al., 2024).

Next, I turn to implications of hold-ups from late worker arrivals at the firm level. I estimate the effect of incentives on total firm output.[4] I find a statistically significant, 0.30 standard deviation increase in weekly firm output – 29% increase relative to the control mean, equivalent to ≈ 700 pieces. Similar to worker level outcomes, gains in total firm output are concentrated among Bottleneck firms, suggesting that the firm also stands to gain from improving coordination in worker arrival times.

Additionally, better coordination in worker arrival times facilitates timely completion of orders. In a context where most firms rely on repeat relationships with clients, timely completion is crucial for sustaining these relationships. Treatment firms complete orders with significantly lower delays – the probability of completing an order after the planned delivery date falls by 10 percentage points, a 40% reduction relative to the control mean. Treatment firms also receive 0.27 SD larger quantity of new orders during the incentive weeks (equivalent to a 21% increase in new orders received relative to control firms, ≈ 850 pieces).

Finally, weekly firm revenues increase by INR 3,257 (8% relative to the control mean), while total labor costs, *excluding* the incentive paid to treatment workers, increase by INR 1,640 (9% relative to the control mean). A simple profit accounting exercise suggests that weekly firm profits, excluding incentive payments made to workers, increased by ≈ INR 1,600 (8% of the control mean). This suggests that absent lateness, firms could yield higher profits. However, paying a 20% bonus as incentives for timely arrival outweighs revenue gains, resulting in a decrease in profits of ≈ INR 981. This is consistent with workers placing a high value on flexible arrival times.

A natural puzzle is why firm owners were not enforcing common start times, or offering other kinds of incentives absent the experiment. At endline, 76% owners were willing to continue if the research team subsidized them.[5] However, only 26% treatment firm owners were willing to continue paying for incentives themselves, and average willingness to pay was INR 30 – less than a third of the incentives that I offered. In a 3-12 month follow-up survey, there was no take-up of any kind of financial incentives among treatment firms. About half of both treatment and control firms reported using soft, verbal encouragement to ask workers to arrive on time; only half of those reported continuing to do so, and a third found such measures to be effective. This lack of enforcement, and low willingness to

[4]Total firm output is measured as the total number of finished pieces produced by all workers in the firm, and standardized to pre-incentive, product-level means.

[5]Anecdotally, treatment firm owners were 26 percentage points (96%) more likely to agree that total production at their firm increased during the incentive weeks, and were twice as likely to attribute this increase to workers arriving on time relative to control firm owners. This indicates that owners did find the incentives valuable.

contract on coordinated arrival times may reflect a broader coordination problem – firms operate in dense clusters with an active market for experienced tailors, and where 95% firms pay tailors piece rates and 80% report they would tolerate lateness by up to 30 minutes. This makes it difficult for a single firm to deviate without facing difficulties in hiring or retaining experienced tailors. Although six weeks is a short time horizon to capture long-term profitability, this paper provides crucial proof of concept that returns to coordination within the firm are large and economically meaningful. The long-run sustainability of such incentives for both workers and firms depends crucially on the state of childcare markets, infrastructure reliability and gender norms around women's participation in household work.

This paper is most closely related to the literature on barriers to firm growth in developing countries. I extend the literature examining the impact of irregular labor supply on firm outcomes (Krishnaswamy, 2019; Goraya et al., 2025) in two key ways. First, this paper provides causal evidence that firms gain from timely worker arrival through coordination gains along the production line using novel, high frequency data on worker labor supply and task assignments within the firm. This extends the finding from previous work that adverse effects of worker absenteeism are stronger in firms and industries with larger co-worker complementarities. Second, this paper sheds light on a novel aspect of labor supply – when workers arrive at work – as a meaningful driver of worker and firm outcomes in developing countries.[6]

A large body of work documents search and hiring frictions in low-income countries (De Mel et al., 2019; Hardy and McCasland, 2023; Bassi and Nansamba, 2022; Caria and Falco, 2024; Carranza et al., 2020). I add to this work by documenting that even conditional on forming matches, firms may face barriers to coordinating workers. Specifically, I show that a lack of coordination in worker arrival times affects both workers' and firms' ability to realize the returns from coordination in a team production setting. To the extent that coordination becomes costlier with scale (Lucas, 1978), this can have important implications for firms' ability to scale their workforce. Additionally, I highlight worker preferences for flexibility as a novel barrier faced by developing country firms in coordinating labor inputs.[7]

[6]Variability in worker arrival times is common in several settings – for example, Cefala et al. (2024) document substantial variability in when workers arrive at labor stands to search for work in Chennai, India. Similarly, in data from 220 micro-enterprises in Coimbatore, India that employ at least 1 paid worker, owners report that all employees showed up at the firm by the expected start time only 2 out of 7 days in the past week. This paper is closest to the work studying tardiness among government workers in developing countries Chaudhury et al. (2006). To the best of my knowledge, this is the first paper to experimentally test the effects of worker tardiness on firm outcomes in the private sector.

[7]As recent work highlights, such preferences may be a consequence of under-development itself, owing to the prevalence of health shocks, poor infrastructure reliability, kinship pressure, etc. (Breza and Kaur, 2025). These reasons are reflected in survey responses from workers in this sample as well: among the top

Finally, I contribute to the growing literature on the internal organization of small, informal firms in developing countries, building on recent work such as Bassi et al. (2022, 2023). In particular, Bassi et al. (2023) find limited labor specialization in multi-worker Ugandan manufacturing firms limits firms' ability to organize labor productively. Instead, these firms are organized as a collection of self-employed individuals, and are unable to realize gains from team production. By contrast, I study a setting with labor specialization along a sequential production line and provide experimental evidence that coordination gains do exist in such a setting.

An additional contribution of this paper is based on the insight that firm owners continue to allow dispersed arrivals, despite gains to both worker and firm output. Since late worker arrivals reflect preferences for flexibility, this paper highlights an important, yet under-explored relationship between the provision of flexible work arrangements by firms, and labor supply preferences (Mas and Pallais, 2020). I experimentally estimate workers' demand for flexible work arrangements and highlight the trade-off between worker preferences for flexibility and firm output. Workers' demand for flexibility has been documented across several different contexts (Mas and Pallais, 2017; Ho et al., 2024; Baysan et al., 2024; Wiswall and Zafar, 2018). This paper's main contribution is to link workers' WTP for flexibility with firm outcomes. I show that while workers individually prefer flexible work arrangements (as demonstrated by (a) the status quo dispersion in arrival times and (b) heterogeneity in response to the incentives), on average, all workers in the firm experience an increase in output and earnings when they arrive by the preferred start time. These findings are consistent with the idea that despite fixed-schedules being a dis-amenity for workers individually, there are potential gains to themselves *and* the firm.

The remainder of the paper is organized as follows: section 2 describes the study context and sample firms, section 3 describes the experimental design, implementation and data collection, and section 4 lays out the empirical strategy. section 5 presents the main results and section 6 concludes.

2 reasons for not arriving on time in the past week, 50% cite household work, followed by 27% citing kids or other household members falling sick. Similarly, when asked why workers prefer flexibility, 45% state household or other personal tasks (such as visiting government offices or ration shops), 23% state childcare or elderly care, 20% state own health shocks, and 16% state needing leaves/time-off for functions or local festivals. Workers also cite long queues at government offices/ration shops and unreliable transportation as reasons they need flexibility for completing personal tasks during the workday.

2 Setting, Data and Descriptive Evidence

2.1 Setting and Data

The context of my study is Tiruppur district in the state of Tamil Nadu, India. This region is part of a major garment manufacturing hub in India, catering to both export and domestic markets since the 1970s. The production process in Tiruppur is highly decentralized and carried out by sub-contracted units, called job-work units – these units will form my study sample. Job work units are *small* factories, employing about 5-15 workers, at least 50% of whom are women.[8]

Firms are organized as batch processing lines, where each worker performs a distinct operation on the garment in batches of 80-100 pieces. Operations need to be performed sequentially in order to produce a finished piece of garment. The main production workers in these firms are tailors, who perform designated tasks along the production line. Firms specialize in broad product categories - such as t-shirts, bottomwear, innerwear, or kids wear. The average weekly order size in these firms is about 4,000 pieces, varying with product type.

The study takes place in 4 neighborhoods located in Tirupur North sub-district, Tirupur, Tamil Nadu. These neighborhoods were identified as high density clusters of small-medium job-work units during prior scoping work and area mapping exercises. I used the following criteria to screen firms to be part of the sample:

1. Has a stitching section
2. Employs at least 2 tailors who are not related to the firm owner or who work for pay
3. Must deliver finished products back to the buyer/retailer
4. Must receive cut-pieces of cloth to be stitched (ie does not only produce own designs)

All firms that satisfy these criteria were eligible for baseline. In case eligible firms were located adjacent to each other, I approached alternate firms in order to minimize spillovers between treatment and control firms located next to each other.

Baseline: I conduct a baseline survey with owners to capture owner and firm characteristics, and a complete worker roster, which is followed by a baseline survey of all workers

[8]While these firms are larger than the typical small developing country firm, they employ a substantial fraction of the workforce within the garment industry in Tiruppur district - as of 2013, the share of total employment in textiles in 10-30 worker firms ranged between 20-30% in rural and urban areas in Tiruppur district (source: Economic Census, 2013-14). Such organization structures are common in many industrial clusters around the world, for example Nadvi and Schmitz (1994); Banerjee and Munshi (2004); Chaudhry and Woodruff (2013); Atkin et al. (2017).

at the firm. Randomization and treatment announcement was done after owner and worker baseline surveys.

Baseline firm and worker characteristics, and balance by treatment arms are reported in Table 1 and Table 2 respectively. On average, firms hire 6 tailors (excluding owners/co-owners), and have been operating for 6-7 years. Firms produce 4,000 pieces of output each week, and earned a profit of INR 30,000 in the last 30 days (USD 340).

Owners directly supervise workers (that is, without a manager/contractor) in almost all firms – owners also perform stitching work in 87% of firms, and work as regular tailors in about 43% of firms. About 85% owners are male, 57% have completed secondary school and have 19 years of experience in the garment industry.

Workers are on average 37 years old (similar to owners), 44% are male and most workers are married (94%) and have at least one child younger than 14 years of age (85%). 87% of workers live within a 15 minute commute to the firm, have worked at the current firm for 2.5 years and have 15 years of experience in the garment industry. Both firm and worker characteristics are balanced across treatment and control firms (Table 2).

Order, Task and Piece Rates Data: At baseline, I also record a roster of all distinct styles that the firm is currently producing. For each style, I record the piece rate that the owner receives for a finished stitched piece and its task structure – a list of all tasks that need to be performed to produce the finished product, the order in which they are performed, and task specific piece rates paid to workers. This roster is updated throughout the 6-week study period to incorporate new styles that the firm is producing.

Daily Production and Attendance Data: Starting 1-2 weeks after baseline, I collect data *daily* for a period of 6-weeks, out of which workers are paid incentives for 4-weeks.[9] Each day, I record workers' attendance, arrival time, break times during the day, and daily output (by product-task). Arrival times and attendance are recorded by surveyor observation and are not prone to mis-reporting in order to game the incentive; remaining data is self-reported by workers. Daily output is recorded for all production workers, including firm owner(s), helpers, and temporary workers where applicable. I use daily output data from all production workers to construct a measure of total firm production. In addition to tailoring/production output, I also record whether the worker did any other work at the firm, such as helper work, receiving or delivering raw material/finished pieces, machine repair, purchasing inputs, etc.

[9]In Phase 1, I collected data 1 week pre-incentives and 1 week post-incentives. In Phase 2, I collected 2 weeks of pre-incentives data only, dropping the post-incentive data collection. In Phase 3, I collected 2 weeks of pre-incentives data, 4 weeks of incentives data and 1 week of post-incentives data.

Orders Data: At the end of each week, I record all orders received and delivered by the firm in the past week. This allows me to track new orders received by firms, defined by new batches of cut-material received, and order completion timelines. Order completion timelines are based on owners' reported planned versus actual completion dates per order, allowing me to construct a measure of production delays.

Endline: Finally, I conduct endline surveys during the last week of the study to capture firm owners' and workers' perceptions of the incentives, preferences for flexible work arrangements, changes in firm operations during the study and time-use during and outside of work (relative to the pre-study period). These questions allow me to examine (1) the nature of workers' commitments/restraints that drive their preference for flexibility, (2) how compliance with the incentive impacted these commitments/restraints, and (3) whether compliance shifted workers' preferences for flexible (versus strict) work arrangements.

2.2 Descriptive Evidence on Worker Arrival Times

Tailors are paid piece rates for each individual task they perform, and not the number of finished pieces of the garment. Although firms have set operating hours, tailors are not explicitly incentivized to arrive at a common start time. As a result, despite the sequential nature of production, the pay contract does not give tailors any private incentive to coordinate their arrival times. Consequently, I observe substantial dispersion in worker arrival times around the owners' preferred start time, as shown in Figure 1: the 10-90 gap of this distribution is 135 minutes – just over 2 hours. For the remainder of the paper, I define late (timely) arrivals as arriving after (on or before) the preferred start time. I use average pre-incentive arrival times to classify workers into three types: "Early", "Late", and "Very Late" workers. "Late" and "very late" are indicators for workers whose average pre-incentive arrival times were between 0 and 30 minutes, and above 30 minutes of the preferred start time, respectively.

Lateness is both persistent and unpredictable. Persistence in timely-arrival across two consecutive workdays for the average worker is 0.55 (SE = 0.04, p-value < 0.001), but falls to 0.06 (SE = 0.041, p-value = 0.175) after accounting for worker fixed effects. About half the variation in timely-arrival is within worker (intra-class correlation = 0.46), suggesting that lateness is a characteristic of workers that can be learned by owners. This is consistent with focus group discussions and survey data from firm owners about which workers at the firm are typically late, and their usual arrival times. However, workers often deviate from their own arrival time norms, making it difficult for owners to predict actual arrival times.

For example, owners could correctly predict which workers will arrive on time the next day 63% of the time, and which workers will arrive on time for the entire week only 23% of the time.[10] On the other hand, owners are better at predicting worker attendance: they correctly predict who will be present the next day 87% of the time, and for the entire week 57% of the time. This motivates understanding the effects of late arrivals on firm outcomes in addition to worker absences.

2.3 How Lateness Affects Production

Lateness can affect production both directly and indirectly. Pre-incentive data on daily work hours and output suggests that "Late" and Very Late workers work fewer hours, but don't produce more pieces per hour relative to Early workers. Thus, regardless of other workers' arrival times, Late and Very Late workers may result in lower firm output through the direct channel of producing fewer pieces of output themselves.

However, given the sequential nature of production, late arrivals may also have indirect effects on other workers in the firm. Specifically, the lack of coordination in arrival times and the presence of both 'Early' and Late/Very Late workers in the firm can create bottlenecks in production in two ways. First, lateness on upstream task stations in the production line can hold-up workers on downstream tasks by either increasing idle time – if early workers wait for the late worker to complete upstream tasks, or limiting downstream workers' task specialization – if early workers need to fill in for the upstream task. Both outcomes are less desirable to workers given workers are not paid on time, and workers prefer to specialize in their primary tasks. On the other hand, lateness on downstream task stations doesn't create externalities on other workers on the production line. However, it may hold-up the delivery of finished garments, and thus create delays in order completion.

I use this stylized framework to define two subsets of my sample to test for indirect effects of reducing lateness within firms. I define a "Bottleneck" firm as a firm with at least 1 upstream task worker who is also "Late" or "Very Late" at baseline. An upstream task worker is a worker whose modal baseline task – indicative of their main task – is an early sequence task in the production line. The complement to this is a "Non-Bottleneck" firm, where no upstream task worker is late. This framework yields two predictions:

1. If lateness creates negative spillovers along the production line, reducing lateness would

[10] Owners are also over-optimistic about timely arrival, and more so for late workers: owners reported 52% of their workers will be on time for the entire week, compared to actual on-time rate of 41%. This gap is larger among workers who are late on average: 43% predicted versus 23% actually on time.

have both direct and indirect effects on worker output. Spillovers should only be present in Bottleneck firms, and not in Non-Bottleneck firms.

2. If lateness does not create negative spillovers along the production line, reducing lateness would only have direct effects on worker output. However, reducing lateness helps reduce order-completion delays.

This framework is consistent with owners' beliefs about how lateness affects production at their firm. At baseline, 98% owners report that they would like their workers to start work at the same time as a worker-management measure that would help increase their firm's production.[11] When asked why coordinated timings would help increase production, 66% of owners suggest a reduction in idle time, and 51% suggest increased task specialization as possible mechanisms (Figure 2b). Additionally, 40% owners also allude to uncertainty about whether and when workers will arrive at the firm as another mechanism. Coordinated timings can help owners plan and deliver orders on time, which helps them receive more orders in the future.

3 Experiment Design and Implementation

Firms are randomly assigned to one of two treatment groups - (T) Treatment (n = 46), and (C) Control (n = 46). All workers in treatment firms were eligible to receive a daily bonus of INR 100-150 (equivalent to 20% of their *average* daily earnings) if they arrived at the firm and started work by the company's preferred start time. Thus, the incentive encourages both on-time arrival and attendance. The total incentive amount was paid to workers as part of their weekly earnings.[12] The unconditional incentive amount for control group firms was matched to the total realized payment made to a randomly selected treatment firm, capped at INR. 300-450 per week (half of the maximum incentive amount in treatment firms). These treatment-control firm (randomized) pairings were created separately each week. Figure 4a shows the experiment design.

The experiment was implemented in three phases - Phase 1 was implemented with 17 firms (9 treatment, 8 control) from November 2024 to January 2025. Phase 2 was implemented with 43 firms (22 treatment, 21 control) from March 2025 to May 2025. Phase 3 was

[11] This far outweighs other measures – 38% of owners would like workers to take fewer/shorter breaks, 25% would like workers to take fewer un-announced leaves and 17% to not bring their children to work (Figure 2a).

[12] All workers who were working full-time at the firm at any point during the study were eligible to receive incentives. Workers' full-time status was determined by the owner and verified by the research team using daily attendance data.

implemented with 36 firms (19 treatment, 18 control) from August 2025 to October 2025. Figure 4b shows the study timeline. The daily treatment incentive amount in Phase 1 and Phase 2 was set to INR 100, and was increased to INR 150 in Phase 3.

Randomization for Phase 1 was conducted by generating a random order of treatment and control assignments, which firms were matched to in the order in which their baseline survey was completed. Randomization for Phases 2 and 3 was stratified on the degree of dispersion in worker arrival times around the owner's preferred work start time, using worker's usual arrival times as reported by the owner at baseline. All specifications reported in the paper control for phase by strata fixed effects to account for the method of randomization in each phase (Bruhn and McKenzie, 2009).

4 Empirical Strategy

I estimate the effect of incentives on worker level outcomes using the following specification:

$$Y_{i,f,t} = \alpha + \boldsymbol{\beta} T_{i,f} \times \mathbb{I}\{\text{During}\}_t + \lambda_{i,f} + \mu_w + \mu_{dow} + \varepsilon_{i,f,t} \tag{1}$$

where $Y_{i,f,t}$ is the outcome of interest for worker i in firm f at time t, and time is either day or week, depending on the outcome variable. $T_{i,f}$ is an indicator that equals 1 if a firm f is assigned to treatment (fixed over time), $\mathbb{I}\{\text{During}\}_t$ is an indicator variable that takes the value 1 if the study week is an incentive week, and $\lambda_{i,f}$ denotes worker fixed effects. μ_w are study-week fixed effects, [13] and μ_{dow} are day-of-week fixed effects (included only in day-level specifications). $\boldsymbol{\beta}$ is the coefficient of interest, and standard errors $\varepsilon_{i,f,t}$ are clustered at the firm level.

I use a similar specification for firm-level outcomes:

$$Y_{f,t} = \alpha + \boldsymbol{\beta} T_f \times \mathbb{I}\{\text{During}\}_t + \lambda_f + \mu_w + \mu_{dow} + \varepsilon_{f,t} \tag{2}$$

where indices and variables are defined as above, and standard errors $\varepsilon_{f,t}$ are clustered

[13] Study weeks take the value 1 to 4 for the 4 incentive weeks, 0 for the week prior to incentives. Weeks 0-4 are common across all phases. The first pre-incentive week in Phases 2 and 3 is coded week -1, and the final, post-incentive week in Phases 1 and 3 is coded week 5. In the current analysis, I drop the post-incentives week (week 5).

at the firm level. Some firm outcomes are only measured at endline, in which case I estimate Equation 2 without the t subscript and without firm and study-week fixed effects, and add the baseline value of Y_f as a control. Day-of-week fixed effects are only included in firm-day level regressions.

5 Results

5.1 Labor Supply

First, I test the effect of the incentives on worker attendance, compliance with work start times and daily work hours, reported in Table 3. Attendance is an indicator for whether the worker was present at the firm that day. Compliance is defined as an indicator for being present at the firm, *and* arriving by the owner's preferred start time.[14] Daily work hours are defined as the total work duration net of the total duration of breaks taken during the day. I find no effect on attendance (column (1)), and an 18 percentage point increase in compliance (column (2)) in treatment firms during the incentives weeks. This represents a 43% increase in compliance relative to the pre-incentive control mean (41 percentage points), and is driven by workers arriving on time, rather than a change in worker attendance. Figure 6 plots the distribution of worker arrival times (conditional on being present) in 15-minute intervals around the preferred start time for treatment (solid lines) and control (dashed lines) firms, visualizing the treatment effect on compliance. Incentives increase the fraction of workers that have arrived within 15 minutes of the preferred start time by 20 percentage points, over a base of 60% in control firms. Given the average size of 6 tailors per firm and 71% attendance rate (4 out of 6 workers present on average), this means that treatment increases number of tailors at the firm by the first 15 minutes of the preferred starting time by 1 additional worker.

Column (3) of Table 3 reports the treatment effect on daily work hours. Daily work hours is defined as the total hours worked in a day, excluding the total time spent on breaks. Breaks include any time taken for lunch, tea, or other reasons for which the worker left the firm.[15] Hours are coded 0 on days that a worker was absent. Workers in treatment firms worked an additional 0.27 hours (16 minutes) relative to workers in control firms, but this effect is not significant at conventional levels.

Next, I examine whether these effects vary by gender, and baseline lateness in Table 4.

[14] Attendance and compliance are both verified by enumerators.

[15] Workers rarely do other non-tailoring tasks for the firm during breaks (5% of worker-days).

Odd (even) columns report the total effect for the omitted group – males ("early" workers) and differential effects for females ("late" and "very late" workers). As shown in columns (1) and (2), treatment effects on attendance for women and "late" workers are no different than their respective base categories, consistent with the result above that incentives did not change worker attendance in treatment firms. Surprisingly, attendance among early workers does not improve, given that these workers were effectively only incentivized to be present at the firm without changing their arrival time.[16] On the other hand, compliance is driven entirely by workers who were consistently late at baseline, and female workers. Female workers are 13 percentage points more likely than male workers to comply (column 3); "late" and "very late" workers are 21 and 35 percentage points respectively more likely than "early" workers to comply (column 4). This is expected, since the incentives were more binding on workers who consistently arrived after the preferred start-time, which was more common among women. Indeed, the increase in work hours is larger for women (0.44 hours, 26 mins) and late workers (0.34 hours/20 mins and 0.73 hours/44 mins respectively), reported in columns (5) and (6) of Table 4, but not statistically significant at conventional levels.

The magnitude of the increase in work hours – 26 minutes for women and 20-44 minutes for "late" and "very late" workers is largely driven by these subgroups of workers arriving earlier than their pre-incentive start times, as opposed to taking longer breaks during the workday or leaving the firm earlier. Table A2 reports the treatment effect on deviation from the preferred start time. On average, workers in treatment firms arrived 11 minutes earlier relative to control workers. This effect is driven by both male and female workers arriving 7 and 14 minutes earlier respectively (column (2)), and "late" and "very late" workers (column (3)) arriving 12 minutes and 27 minutes earlier relative to control workers.[17]

[16] 37% of worker absences are because of social functions, followed by 19% because there was no work at the firm, 18% because either the worker or a family member was sick, and 10% because of local temple/religious festivals. A majority of these reasons relate to importance of social/kinship relations, as emphasized in Breza and Kaur (2025), but it is worth noting a large fraction of absenteeism attributable to low labor demand. This matches the rate of all-day firm closures recorded in the data - 10% of firms were closed for the entire day. 40% of closures were because the firm did not have work, followed by 36% because of local temple festivals or the firm owner's personal commitments, and 10% because of power cuts.

[17] There is some evidence that "very late" workers compensate for arriving earlier by taking one additional break during the day, resulting in an 8 minute increase in the total duration of breaks taken during the day (pval = 0.10), as reported in columns (3) and (6) of Table A4. Finally, Table A3 suggests there is no evidence that workers leave the firm before 6.30[18] (columns (1)-(3)), before the firm's usual closing time (columns (4)-(6)), or before their own usual exit time (columns (7)-(9)).

5.1.1 Does treatment improve worker coordination in arrival times?

Next, I examine how treatment affects worker coordination at the *firm-day level*, conditional on being present. Table A5 reports treatment effects for three measures of dispersion in worker arrival times at the firm-day level: minutes between the 1st and last worker to arrive at the firm (column (1)), the 90-10 gap in worker arrival times (column (2)), and the deviation from preferred start time for the 90th percentile worker (columns (3)). This regression excludes firm-days on which only 1 worker was present at the firm, since the notion of coordination is not meaningful in such cases.

Total dispersion in worker arrival times is 129 minutes on average at baseline – treatment reduces dispersion by 22 minutes, a 17% reduction in dispersion, or improvement in coordination relative to control firms (column (1)). A more conservative measure that excludes extreme outliers - the 90-10 gap in arrival times – also reduces by 9.5 minutes (p-value $<$ 0.1) in Treatment firms, an 11% decrease relative to the control mean of 84 minutes. Finally, I look at treatment effects on deviation from the owner's preferred start time for the 90th percentile worker. Note that given the average size of firms (6 tailors), and assuming 71% attendance, the 90th percentile worker is equivalent to the last or second to last worker to arrive at the firm. This measure also tests whether these effects are driven by workers for whom the incentive was binding. The deviation of the 90th percentile worker reduces by 14 minutes (p-value $<$ 0.01) in treatment firms, relative to the control group mean of 47 minutes pre-incentives. These effects amount to a 29% reduction in dispersion, or improvement in coordination as a result of treatment.

5.2 Worker Output and Earnings

Next, I examine the effect of incentives on worker output and earnings. Worker output is measured as the total quantity of pieces stitched by the worker in a week, across all tasks (and products). This measure does not necessarily correspond to firm output (total number of finished pieces), since a worker who stitched 100 collars and 100 sleeves would be considered to have worked on 200 pieces on that day. However, this measure aligns best with workers earnings, which are constructed as the product of the task specific piece rate and quantity produced on that task, summed over all tasks and products worked on in a week. Output and earnings are coded as 0 on days that a worker was absent, before aggregating to the weekly level. Since firms produce different types of products, I normalize weekly worker output to pre-incentive, product-level means and standard deviations to account for any level differences in task speed and production quantity across product types. I report effects

on worker earnings both excluding and including treatment incentives.

Worker Output Results for weekly worker output are reported in columns (1)-(3) of Table 5. The treatment effect on weekly worker output is 0.27 standard deviations and statistically significant at the 5% level. This amounts to a 24% increase in weekly output relative to the control group mean (column (1) of Table 5). This increase is driven equally by male and female workers (column (2)), and early and late workers (column (3)). In levels, this amounts to 900 additional pieces per week in treatment firms over an average of 3,700 pieces per week in control firms.

Worker Earnings Results for weekly worker earnings *excluding* incentive payments are reported in columns (4)-(6) of Table 5. I report total worker earnings over the week by summing the product of task-specific output and piece rates over the week, assuming that piece rates account for level differences across products. The treatment effect on worker earnings mirrors the results on worker output, albeit smaller in magnitude and not statistically significant. Weekly earnings increase by INR 310 (9% of the control mean) for the average worker. The effects are larger among female workers (INR 424, 12% of the control mean, p-val = 0.11), but similar across early and "late" workers. *Including* the treatment incentive payments, all workers in treatment firms earn INR 817 per week (see Table A6). The increase in earnings including incentives is experienced by all workers (by design of the incentives), but is the largest among female workers (INR 914). These results suggest that if workers were to, hypothetically, arrive on time *without* any monetary incentives, they could have earned on average INR 310-424 more per week than their status quo arrival times. Depending on whether workers internalize indirect effects of timely arrival along the production line, the implied willingness to pay ranges between INR 310 to INR 817 in weekly earnings.

5.2.1 Evidence for coordination gains along the production line

Notably, unlike the results on worker compliance, worker output and earnings are not only concentrated among late and very late workers, indicating the presence of positive spillovers within the firm as a result of treatment. Do these spillovers reflect gains along the production line? To assess this further, I examine whether spillover effects on early workers are concentrated among Bottleneck firms – firms where late workers performed upstream tasks at baseline. If the increase in worker output reflects gains from coordination along the production line, spillovers to workers for whom the incentives were not binding – early workers – should appear only in firms where late arrivals actually held up production. Table 6 replicates results from Table 5 separately by bottleneck status. Panel A reports results for

52 Bottleneck firms, and panel B reports results for 40 Non-Bottleneck firms. The overall effect on weekly output, reported in column (1), is larger in Bottleneck firms – 0.36 standard deviations (32% of the control mean, significant at the 1% level) in Bottleneck firms, compared to 0.12 standard deviations (12%, not statistically significant) in Non-Bottleneck firms (column (1) of Panels A and B). More importantly, which workers gain from the incentives differs across the two samples. Column (2) reports heterogeneity in treatment effects by early and late status. Panel A reflects the same pattern as column (3) of Table 5 – all workers, *including* early workers gain from the treatment. All workers experience a 0.39 SD increase in weekly output (equivalent to 1,000 pieces), significant at the 5% level. Total effects for late and very late workers are similar in magnitude (0.37 SD and 0.32 SD respectively) and are not statistically distinguishable from the effect for early workers. By contrast in Non-Bottleneck firms (Panel B), the effect is concentrated entirely among late workers – late (very-late) workers experience a 0.26 (0.47 SD) increase in weekly output ($\approx$ 1,500-2,000 additional pieces), although only the effect for very late workers is statistically significant at the 10% level (p-values of 0.32 and 0.08 respectively). Weekly worker earnings (columns (3) and (4)) exhibit the same pattern – all workers in Bottleneck firms experience an increase in earnings, although the estimate is not precisely estimated. On the other hand, in Non-Bottleneck firms, late and very-late workers experience an increase in earnings of INR 540–550 (19% of the control mean), although neither effect is statistically significant (p-values of 0.35 and 0.14 respectively).

Taken together, spillovers to workers for whom the incentives were not binding appear only in firms where an upstream worker was late at baseline. This is what a coordination mechanism predicts, and is difficult to reconcile with alternative explanations for the output gains, such as peer effects from arriving together, or morning hours simply being more productive – both would generate gains among early workers in Bottleneck and Non-Bottleneck firms alike.

5.2.2 Worker output per hour

To what extent are the increases in worker output simply mechanical – i.e., resulting from an increase in work hours at the firm, versus reflecting productivity improvements? To investigate this, and to provide further support for the existence of coordination externalities, I estimate the effects of treatment on worker output per hour, reported in Table 7. For the average worker, output per hour increases by 4.5 units per hour, a 5% increase relative to a control mean of 87 units per hour (column (1)). Consistent with the results on total weekly worker output, this increase is realized by both male/female (column (2)), and early/late

workers (column (3)). However, "late" workers experience the largest increase in output per hour - 7.62 units (p-value = 0.09), equivalent to 9% of the control mean. This increase is consistent with "late" workers being less likely to perform non-production tasks at the firm when they are present, as shown in column (3) of Table A7. "Late" and female workers are 5 percentage points less likely to perform non-production tasks[19], equivalent to a 33% reduction relative to the control mean of 16 percentage points. Although not statistically significant, this reduction is economically meaningful to the firm for several reasons. First, this indicates that a subset of tailors are able to specialize in tasks they have a comparative advantage in - tailoring. Second, anecdotally, owners report that tailors typically perform helper tasks when orders are urgent (50%), helpers are absent (33%), there is no other work at the firm (30%), or when tailors have idle time (14%). Table A8 shows that there is no change in helper attendance in treatment firms during the incentive weeks. At endline, treatment firm owners are more likely than control owners to report that both tailors and helpers experienced less idle time at the firm during the incentive period (Figure A1). Together these results provide evidence in support of better time allocation among workers in treatment firms.

5.3 Total Firm Output

Finally, how do the changes in individual worker output add up to total firm output and revenues? I construct total firm output using daily production data collected from workers by aggregating total quantity produced in two ways: (1) across *all* tasks, and (2) for the last task in the production line. These measures correspond to total units stitched in the firm, and total units of the finished garment respectively. Firm revenues and total labor costs are computed as the product of piece rates earned by owners and workers, and the total quantity of the last task in the production line. Both measures aggregate output across all production workers (i.e., including production by tailors, firm owners and managers, helpers and temporary workers). Treatment effects are estimated using Equation 2 and reported in Panel A of Table 8.

Total firm output increases by 0.30 standard deviations (column (1)) – this is equivalent to 6,800 additional pieces stitched at the firm, a 29% increase relative to the control mean and is statistically significant at the 10% level. The last task measure, reported in column (2) increases by 0.21 standard deviations, equivalent to 700 additional pieces of finished output. This measure is noisier, perhaps because the last task is under-reported if orders are still under production. Panel A of Table A11 reports these estimates after dropping the last week

[19]Non-production tasks include helper work, and other work outside the firm like receiving new material, making deliveries, purchasing thread, etc. Most non-production tasks are helper tasks.

of data collection, and the magnitudes across both measures are similar. Reassuringly, a third measure counting all but the last task on the production line (column (3) of Table A11A) produces similar estimates. Finally, consistent with results on worker output, all of the gains in total firm output are concentrated among Bottleneck firms, as reported in Column (1) of Table 8B. Total weekly output increased by 0.46 SD (41% relative to the control mean, p-value = 0.02) in Bottleneck firms, while the effect in Non-Bottleneck firms is 0.13 SD (15% relative to the control mean, p-value = 0.59). However, I fail to reject the difference between the two samples.

An alternate measure of total firm output is the quantity of new orders received by firms, reported in column (3) of Table 8. This measure was collected directly by firm owners and serves both as an independent measure of firm output[20], and firms' ability to procure new orders. Treatment firms procure 0.27 standard deviations more per week in new orders, a 21% increase relative to the control mean ($\approx$ 850 pieces). This measure is noisily estimated, but the magnitudes match those from daily production data collected from workers. Unlike the results on total firm output, both Bottleneck and Non-Bottleneck firms receive a larger quantity of new orders. However, both estimates are noisy and I am unable to reject the difference between both samples, making these results suggestive.

Finally, I find that treatment firms are more likely to deliver orders on time relative to control firms. Figure 7 reports treatment effects on order completion delays, defined as an indicator of whether the difference between planned and actual days of delivery exceeds 1 day. Treatment firms complete orders with significantly lower delays – the probability of completing an order after the planned delivery date falls by 10 percentage points, a 40% reduction relative to the control mean. Better coordination in worker arrival times facilitates timely completion of orders. In a context where most firms rely on repeat relationships with clients, timely completion is crucial for sustaining these relationships.

5.4 Profitability of Incentives

Total weekly firm revenues (column (1) of Table 9) in treatment firms increased by INR 3,257, an 8% increase relative to the control mean (INR 39,000), and total labor costs, *excluding* the incentives increase by INR 1,640, a 9% increase relative to the control mean (INR 17,843). However, both effects are noisy, and not statistically significant at conventional levels. Finally, total labor costs including the incentives paid to workers (column (3)) increase by INR 4,239 (24% relative to control firms).

[20] 98% new orders were delivered, so this indicates total pieces delivered by the firm as well.

These results suggest that improving worker coordination increases firm output, increases both firm revenues and labor costs (although noisily estimated). However, increases in labor costs including incentives outweigh the gain in revenues, resulting in a reduction in profits by INR 981 (5% of the control mean, reported in column (6)). Even including incentives paid to the marginal workers only (i.e., "Late" and "Very Late" workers) – while not a feasible contract – firms would just break even (column (7)). Thus, while the lack of coordination and lateness in worker arrival times results in large loss in production, it is even costlier for firms to incentivize timely arrival. However, these calculations don't rule out the existence of any contract that could incentivize timely arrival. The choice of 20% daily bonus in this paper was deliberate in maximizing the first stage response from workers.

5.5 Why were owners not incentivizing timely arrival before the experiment?

A natural question arises – if lateness is indeed harmful to firm productivity, why do owners allow lateness in the status quo? I provide some suggestive reasons based on survey responses and anecdotes from the field.

First, the existing equilibrium in which these firms operate makes it hard for a single firm to enforce common work start times without raising wages or incentivizing workers. Firms are located in dense clusters, where over 95% of firms pay workers on piece rate contracts, and 80% owners report being okay with workers arriving 30 or more minutes late. When asked about the consequences of enforcing common work start times without incentives, owners believe that workers may leave the firm/refuse to work at the firm (52%), or they may be less willing to work extra hours when there are urgent orders (38%), and that it may harm their relationship with workers (31%). These reasons point to a combination of hiring frictions, idiosyncratic nature of demand and reputation concerns for owners.

Second, owners may have incorrect beliefs about workers' response to such incentives, or incorrect beliefs about the profitability of such incentives. In endline surveys, treatment firm owners are more likely to agree that their firm's total production and profits increased during the incentive weeks. This suggests that the experiment itself may have corrected owners' beliefs about the viability of such incentives. 76% of treatment owners reported that they would be willing to continue the incentives if the research team continued to pay for them, indicating that most owners found the incentives were useful. However, only 26% reported that they would be willing to continue the incentives by paying themselves. On average, treatment owners say that they would be willing to contribute INR 30 per day out of INR

100 for such incentives, and 19 out of 40 owners reported 0, indicating a low willingness to pay.

Finally, it may in fact be too costly to incentivize workers in the long-run. While this paper demonstrates the possibility of gains in the short-run, it is not clear whether workers may continue to arrive on time if incentives were to continue indefinitely. The long-run response of workers depends on several other factors. In particular, missing or thin markets for services such as childcare, transportation (public transit, school buses), or the quality of public infrastructure (municipal water supply, regularity of government ration shops) affect workers' time-use outside of work, and are a key determinant of long-run responses. Perhaps low reported willingness to pay of the owners reflects their beliefs about long-run responses from workers.

Another important long-run consideration is contracting frictions. 7 out of 40 treatment firm owners said that they would be willing to contribute the full amount of the incentive that I offered. However, in conversations with owners after the endline survey, many expressed concern about being able to implement such incentives themselves. Their reasons ranged from contracting frictions to concerns about fairness within the firm. For example, one owner said: "*I'm willing to offer up to INR. 200, but if workers get used to it, they may start considering it as part of their salary and still not come on time.*" Another said "*workers did not feel upset because the incentive came from a third party. If I implement the same system, workers would perceive it as partiality.*" The existence of contracting frictions in developing countries is well-known, and is key to understanding why owners were not implementing such incentives before the experiment.

6 Conclusion

This paper experimentally tests whether irregular labor supply – specifically, a lack of co-ordination in worker arrival times – affects firm productivity in the context of a garment manufacturing cluster in India. I document that workers exercise considerable flexibility in when they arrive at work – at baseline, the 90-10 gap in worker arrival times is over 2 hours. This stands in contrast to firm owners' preference to have workers arrive at a common time, given the production line based organization of firms.

A daily bonus equivalent to 20% of worker earnings successfully increases timely arrival in Treatment firms by 43%, resulting in a 17% decrease in dispersion in arrival times at the firm level. This is a meaningful effect given the small size of these firms – incentives increase the fraction of workers who arrive within the first 15 minutes of opening time from 60% (3

out of 5 workers) to 80% (4 out of 5 workers).

As a result of the incentives, weekly output per worker increases by 24% for the average worker. This partly reflects an increase in productivity (output per worker-hour), given the smaller increase in weekly work hours. More importantly, this increase is distributed equally among "Early" and "Late" workers in treatment firms, indicating the presence of positive spillovers since the timely arrival response was driven entirely by "Late" workers. These spillovers are present in firms where lateness creates the largest hold-ups along the production line, thus reflecting gains from better coordination in production. In firms with little hold-ups along the production line, gains are realized through a reduction in order completion delays – an equally crucial margin that helps firm owners maintain important client relationships. These gains translate almost one-to-one to total firm output, which increases by 29%. Average weekly revenues increase by 8% in treatment firms, and accounting for labor costs without incentives yields 8% higher profits. These results suggest that lateness does negatively affect firm productivity.

However, the costs involved in coordinating worker arrival times are too costly, and not profitable for firm owners. Perhaps not surprisingly, only 26% owners said they would be willing to offer such incentives themselves, and the average stated willingness to pay for a similar daily bonus was INR 30 – less than a third of the incentives that I offered in the experiment, indicating low willingness to pay. In a 3-12 month follow-up survey, I find no take-up of any kind of financial incentives by Treatment firms, and little effort overall in enforcing timely arrival.

Survey responses and anecdotal evidence suggest several possible constraints explaining why willingness to pay is low, and why owners were not implementing such incentives in the status quo. These reasons range from incorrect information and beliefs to norms, missing markets and contracting frictions that are prevalent in many low-middle income contexts. The latter three may require larger scale policy efforts to shift owners' willingness to change the types of contracts and incentives offered to workers. My results demonstrate the importance of policies aimed at improving supply of and access to childcare, and improving the quality of public services such as transportation and municipal residential services (e.g. water supply) as complementary to industrial policies. This also motivates the need for future research to understand what drives the nature of employment contracts offered by firms in developing countries, and their effects on firm productivity.

Finally, it is also striking that incentives did not change worker attendance, and that take-up does not increase to 100% workers arriving on time conditional on attendance. This

is despite the incentives being 20% of average daily earnings, indicating that some workers very highly value flexibility. In other survey work with workers across Tirupur district, I document that the provision of flexibility varies systematically with firm size – small and medium firms are significantly more likely than large firms to offer their workers flexibility in work schedules. Consequently, there is sorting of workers into small ($\leq$ 20 workers) versus large ($>$ 20 workers) firms along gender, marital status and having children – characteristics that are positively correlated with demand for flexibility. While workers who work in small firms enjoy flexible work schedules, they do so at the cost of variable pay and lack of access to formal government benefits, such as state provided health insurance, social security, etc. To the extent that demand for flexibility affects workers' labor supply choices, such as whether to be self-employed, work as casual day labor (Oh et al., 2024; Cefala et al., 2024), or work in small firms without formal benefits, this can have important implications for the rate of wage-employment in developing countries. More research is needed to understand the drivers of the demand for flexibility in developing countries, as well as the broader economic implications on the functioning of labor markets and firms in such settings.

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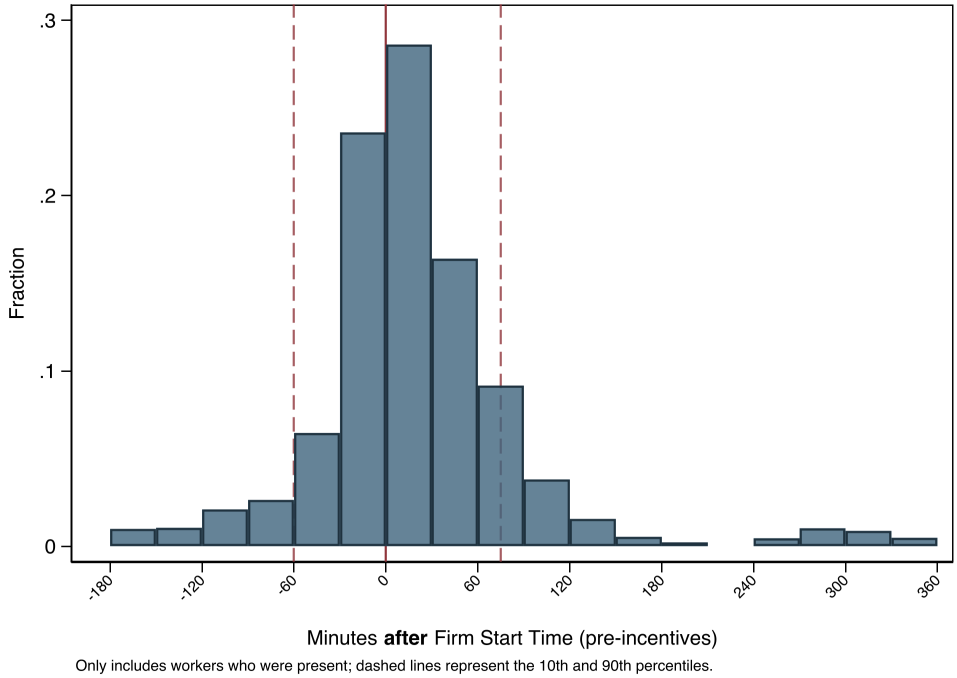
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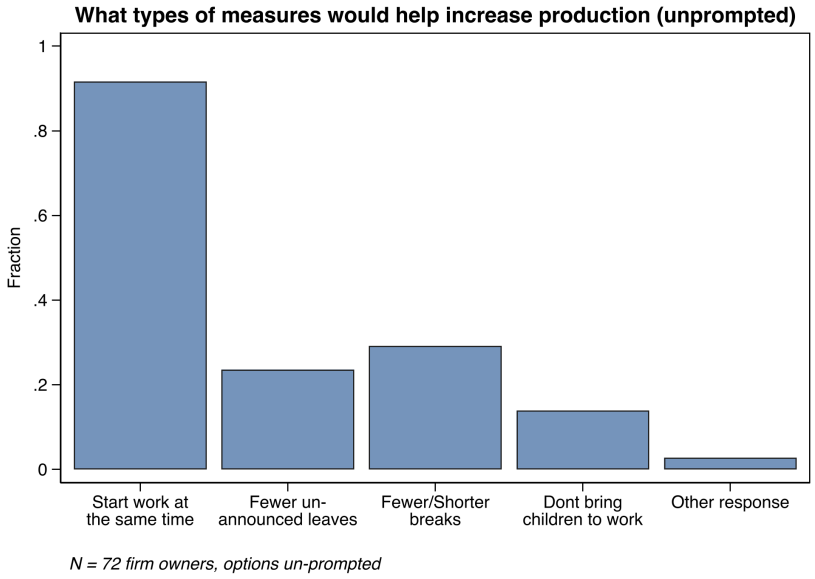
7 Figures

Figure 1: Dispersion in Worker Arrival Times, before Incentives

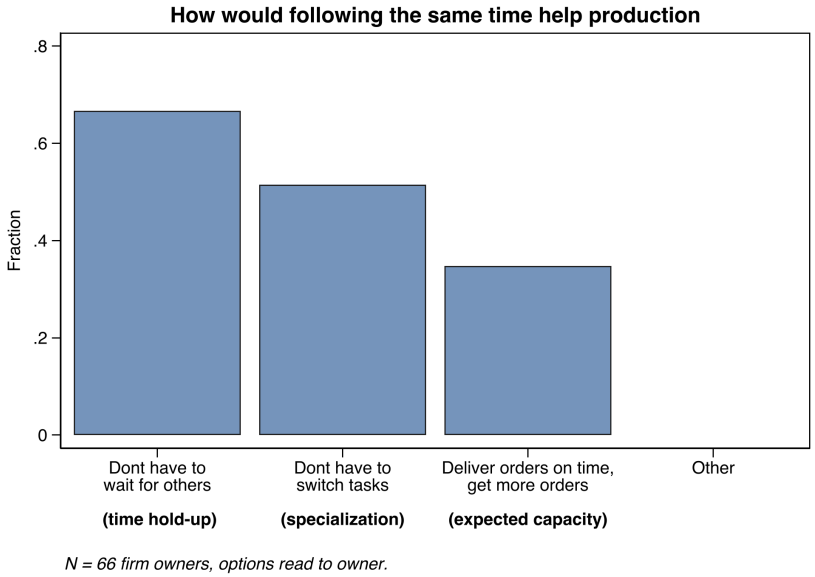


Notes: This figure plots the distribution of the deviation of worker arrival times from the owner's preferred start time. N = 92 firms, ≈ 600 workers.

Figure 2: Importance of Common Arrival Times (Baseline)



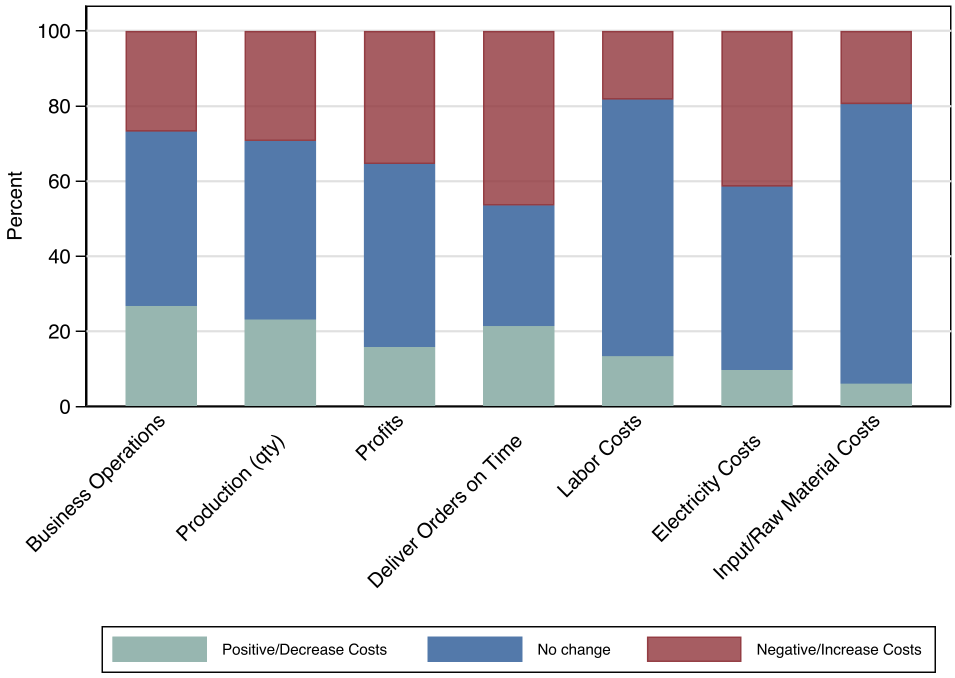
(a) Measures that could help increase production



(b) Why is coordination in worker arrival times important

Notes: Panel A reports owner's responses from the question "*What type of measures would help increase production?*", with options unprompted. N = 71. Panel B reports owner's responses from the question "*If everybody starts work at the same time, how will it improve production?*", with options 1 = Workers don't have to wait for others, 2 = Workers don't have to switch tasks, 3 = Helps me fulfill orders on time and get more orders, and 4 = Others.

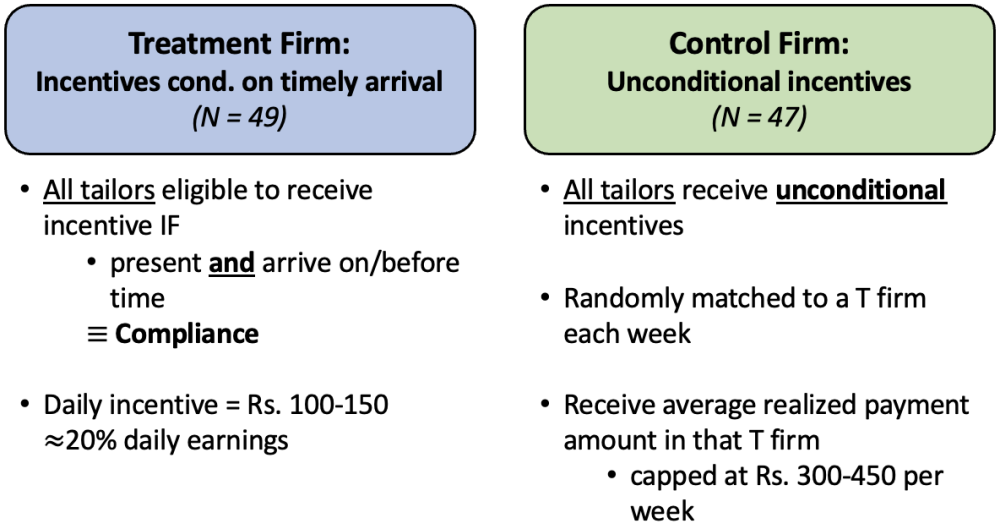
Figure 3: How Providing Flexibility Affects Firm Operations



Notes: This figure reports owner's responses from the question: "*Now I will list 4 different business objectives. For each of these, do you think providing workers flexibility affects this objective positively, negatively, or has no effect?*" This data is from the full sample of 92 firm owners.

Figure 4: Experimental Design and Implementation Timeline

(a) Experiment Design



(b) Study Timeline

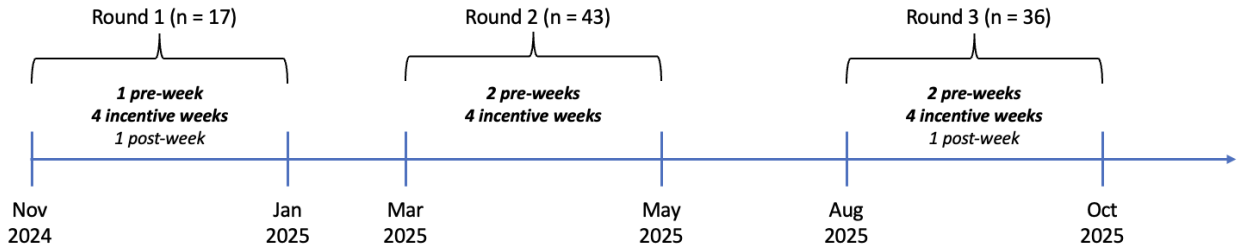


Figure 5: Change in deviation from owner's preferred start time

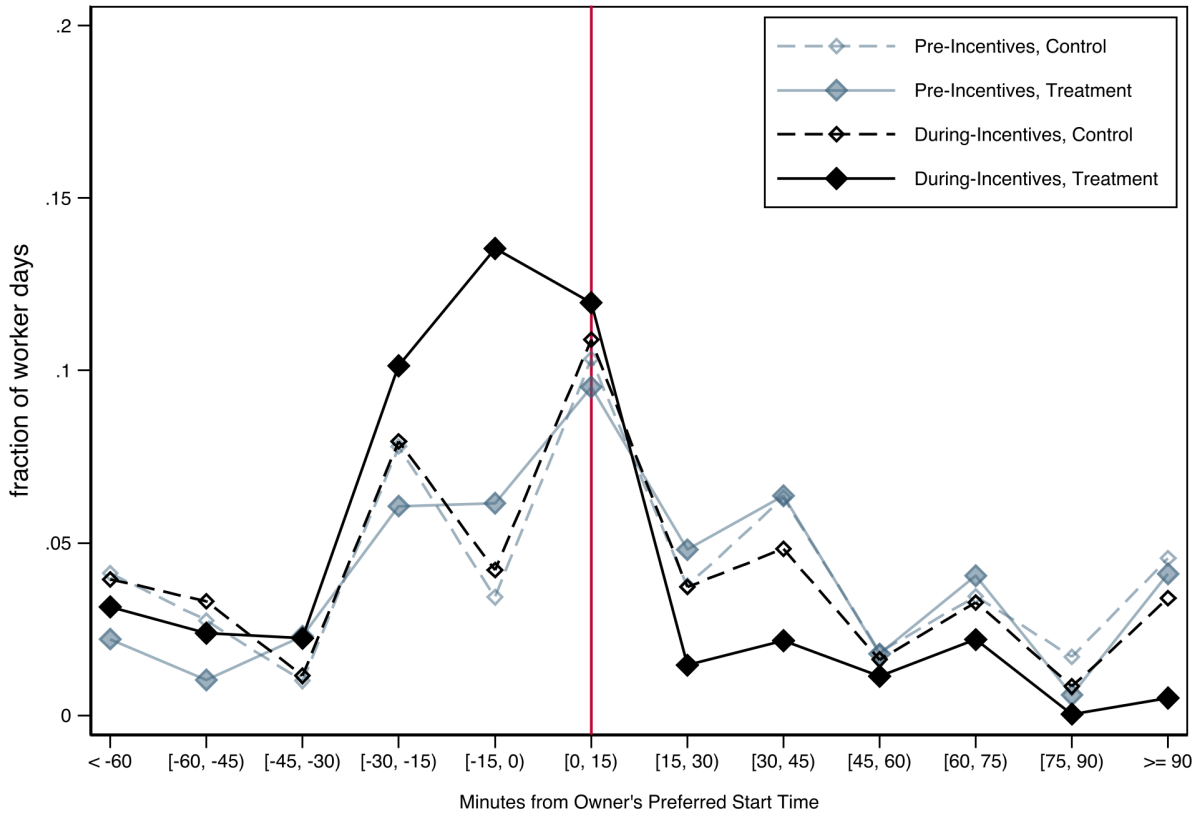
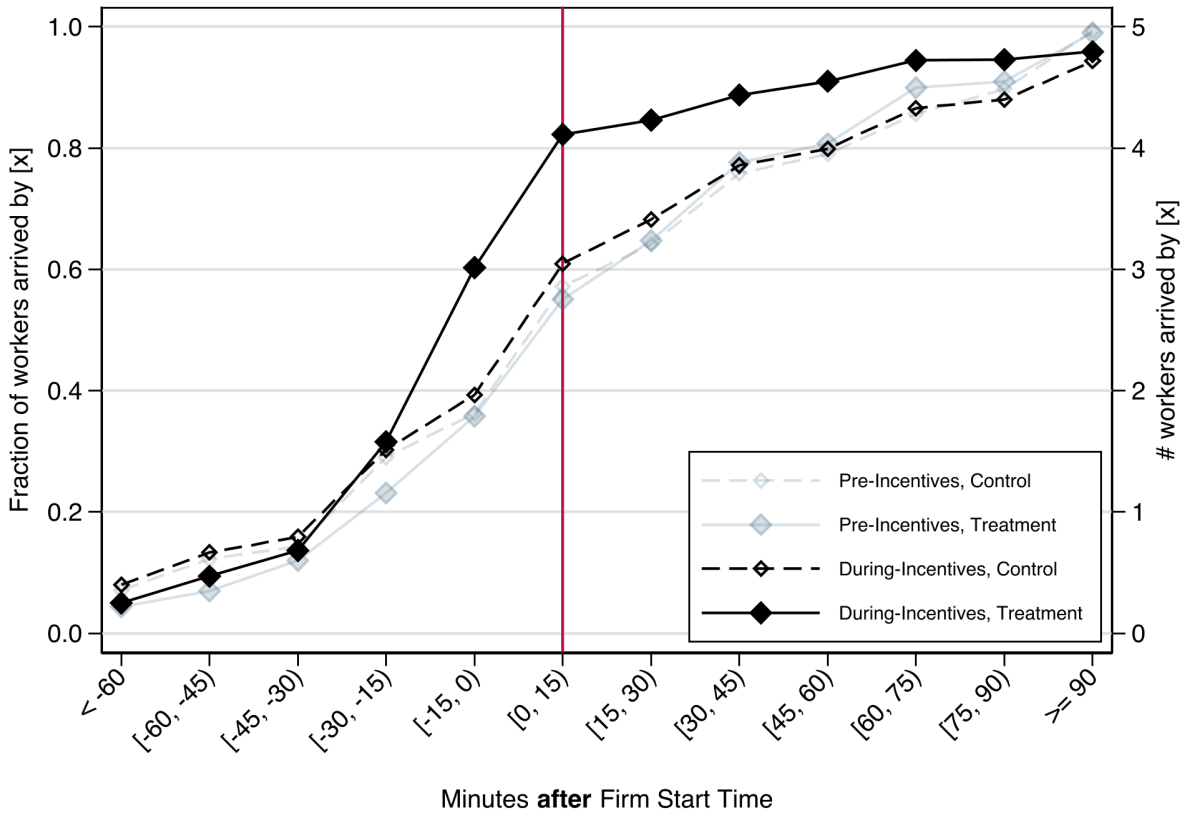
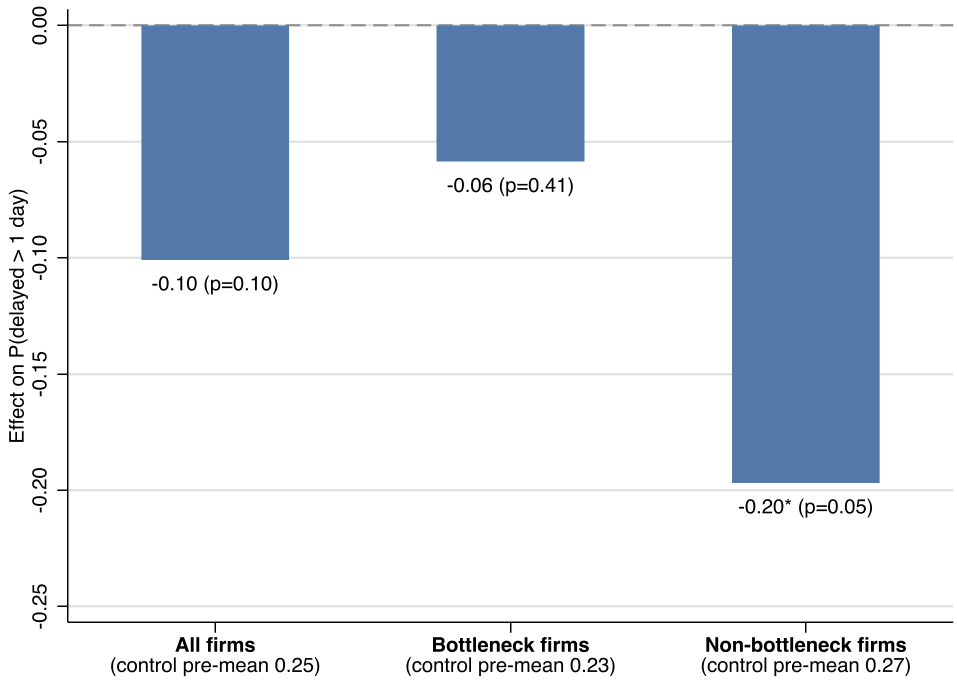


Figure 6: CDF of fraction of workers arriving in 15-minute intervals



Faint blue lines are pre-incentive means

Figure 7: Treatment Effects on Order Completion Delays



Notes: This figure plots treatment effects on the probability that an order is delivered more than 1 day after its planned delivery date, for all firms and separately for Bottleneck and Non-Bottleneck firms. Labels report point estimates and p-values; control means are pre-incentive. * $p < 0.10$.

8 Tables

Table 1: Balance on Baseline Firm Characteristics

	Control	Treatment	T - C	*p-value*
	Mean/SD	Mean/SD	Difference	
Firm Characteristics				
# Tailors	6.47	6.22	-0.18	0.75
	(2.93)	(2.90)		
# Helpers	1.64	1.29	-0.37	0.12
	(1.15)	(1.15)		
Total # Machines	11.64	11.06	-0.44	0.59
	(4.86)	(3.63)		
Owner directly supervises workers	0.91	0.98	0.06	0.20
	(0.28)	(0.14)		
Firm Age (yrs)	6.68	6.76	0.05	0.97
	(5.29)	(6.18)		
Export orders	0.13	0.16	0.04	0.61
	(0.34)	(0.37)		
Only stitching section	0.57	0.63	0.05	0.59
	(0.50)	(0.49)		
Weekly Output	4075.61	4306.98	286.90	0.59
	(2320.43)	(2876.29)		
Revenues (last 30 days)	134444.19	126900.00	-8054.10	0.70
	(118877.68)	(101095.00)		
Profits (last 30 days)	33962.98	25916.67	-8173.26	0.36
	(58396.33)	(18038.77)		
Total Costs - wage bill (last 30 days)	32478.68	32978.37	842.78	0.83
	(20378.58)	(18684.70)		
Owner Demographics				
Age	39.21	37.80	-1.28	0.44
	(8.36)	(8.77)		
Male	0.85	0.82	-0.03	0.73
	(0.36)	(0.39)		
Completed Secondary School	0.57	0.55	-0.01	0.88
	(0.50)	(0.50)		
Commute Time (mins)	6.32	8.63	2.38	0.15
	(6.66)	(9.06)		
Experience in Industry (yrs)	19.15	18.35	-0.91	0.65
	(10.47)	(9.50)		
Owner does stiching work	0.87	0.88	0.00	1.00
	(0.34)	(0.33)		
Owner stitches as a regular tailor	0.43	0.43	-0.01	0.95
	(0.50)	(0.50)		
Observations	47	49		
Joint F-test p-value				0.475

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with robust standard errors. The joint F-test p-value is from a regression of treatment on all balance variables with strata fixed effects and robust SEs; missing values in weekly output, revenue, and profits are mean-imputed within strata for this test only.

Table 2: Balance on Baseline Worker Characteristics

	Control	Treatment	T - C	*p-value*
	Mean/SD	Mean/SD	Difference	
Age	37.22	36.49	-0.78	0.40
	(8.75)	(9.01)		
Male	0.45	0.43	-0.02	0.73
	(0.50)	(0.50)		
Married	0.87	0.84	-0.03	0.41
	(0.34)	(0.37)		
Has children $<$ 14 yrs	0.89	0.84	-0.05	0.72
	(1.26)	(1.16)		
Has elderly member in HH	0.36	0.40	0.04	0.41
	(0.48)	(0.49)		
Related to owner	0.21	0.21	0.00	0.99
	(0.41)	(0.41)		
Native to TN	0.92	0.95	0.03	0.24
	(0.27)	(0.21)		
Commute $<$ 15 mins	0.85	0.83	-0.02	0.48
	(0.36)	(0.38)		
Completed Secondary School	0.46	0.41	-0.05	0.26
	(0.50)	(0.49)		
Experience in Industry (yrs)	15.52	14.15	-1.43	0.14
	(9.25)	(9.53)		
Tenure at Firm (yrs)	2.57	2.38	-0.17	0.73
	(3.60)	(3.20)		
Observations	309	299		
Joint F-test p-value				0.340

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with standard errors clustered at the firm level. The joint F-test p-value is from a regression of treatment on all balance variables with strata fixed effects and SEs clustered at the firm level.

Table 3: Worker Attendance and Compliance

	Timely Arrival	**Daily Work Hours**	**Present**
	(1)	(2)	(3)
Treatment × During Incentives	0.18***	0.28	0.02
	[0.03]	[0.29]	[0.03]
Control Mean (Pre-Incentives)	.41	6.45	.71
Control Mean (Incentive Weeks)	.44	6.85	.68
Treatment Effect %	43	4	3
Worker, Week FE	X	X	X
Day-of-week FE	X	X	X
Worker-Days	17583	16548	17585

Notes: All specifications include worker, day-of-week, phase by week, phase by strata fixed effects. Standard errors are clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Timely arrival is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Heterogeneity in Treatment Effects on Compliance and Attendance

	Timely Arrival		**Work Hours**		**Attendance**	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	0.10**	0.01	0.07	-0.05	0.01	0.02
	[0.04]	[0.04]	[0.32]	[0.42]	[0.03]	[0.04]
Treatment × During × Female	0.13***		0.37		0.01	
	[0.04]		[0.33]		[0.04]	
Treatment × During × Late		0.21***		0.42		-0.01
		[0.05]		[0.47]		[0.05]
Treatment × During × Very Late		0.35***		0.78		0.00
		[0.06]		[0.51]		[0.05]
Control Mean (Pre-Incentives)	.41	.41	6.45	6.45	.71	.71
Omitted Group Mean (Pre-Incentives)	.55	.69	6.76	7.42	.71	.78
Total Effect (T × During + Inter.)	0.23	0.22, 0.36	0.44	0.37, 0.73	0.02	0.02, 0.03
p-val Total Effect = 0	0.00	0.00, 0.00	0.21	0.26, 0.12	0.47	0.65, 0.55
Treatment Effect %	57	54, 88	7	6, 11	3	2, 4
Worker, Week FE	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X
N Worker-Days	17583	15852	16548	15272	17585	15853

Notes: All specifications include worker, day-of-week, phase by week, phase by strata fixed effects, from the specification in Equation 1. Standard errors at clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Timely arrival is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.
Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Weekly Output and Earnings

	Weekly Output (SD)			**Weekly Earnings (INR)**		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	0.26**	0.25*	0.22	310.15	164.62	349.08
	[0.11]	[0.13]	[0.14]	[271.34]	[345.07]	[372.16]
Treatment × During × Female		0.01			259.67	
		[0.10]			[261.11]	
Treatment × During × Late			0.06			-172.40
			[0.13]			[387.26]
Treatment × During × Very Late			0.06			52.80
			[0.14]			[332.68]
Control Mean (Pre-Incentives)	3873.79	3873.79	3873.79	3505.3	3505.3	3505.3
Control SD (Pre-Incentives)	3582.97	3582.97	3582.97	3073.26		
Omitted Group Mean (Pre-Incentives)		3818.15	3847.45		4097.95	4042.68
Total Effect (T × During + Inter.)		0.26	0.28, 0.28		424.28	176.68, 401.88
p-val Total Effect = 0		0.02	0.03, 0.03		0.11	0.60, 0.13
Treatment Effect %	24	24	26, 26	9	12	5, 11
N Worker-Weeks	2663	2663	2602	2663	2663	2602

Notes: SEs clustered at firm level. All specifications include worker, phase by week, phase by strata fixed effects. This regression excludes (a) owners/co-owners who also do stitching work at the firm, and (b) days of firm closure if no production took place at the firm. Output is normalized by product relative to pre-incentive week means. Earnings are coded as 0 for absent workers. Output and earnings are winsorized at the 99th percentile. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Worker Output and Earnings: Bottleneck Spillovers

	Weekly Output (SD)		**Weekly Earnings (INR)**	
	Overall	Lateness	Overall	Lateness
	(1)	(2)	(3)	(4)
Panel A: Bottleneck Firms				
Treatment × During	0.36***	0.39**	525.22	744.60
	[0.13]	[0.19]	[350.27]	[514.06]
× Late		-0.03		-453.00
		[0.17]		[504.61]
× Very Late		-0.07		-159.59
		[0.19]		[446.36]
Control Mean (Pre-Incentives)	3828.82	3828.82	3946.21	3946.21
Omitted Group Mean (Pre-Incentives)		4337.28		5389.93
Total Effect (T × During + Inter.)		0.37, 0.32		291.60, 585.01
p-val Total Effect = 0		0.01, 0.03		0.47, 0.09
Treatment Effect %	32	33, 29	13	7, 15
N Worker-Weeks	1728	1698	1728	1698
N Firms	52	52	52	52
Panel B: Non-Bottleneck Firms				
Treatment × During	0.12	-0.05	112.03	-250.70
	[0.19]	[0.19]	[410.20]	[495.18]
× Late		0.31		792.58
		[0.24]		[622.04]
× Very Late		0.52**		800.36*
		[0.22]		[440.50]
Control Mean (Pre-Incentives)	3933.42	3933.42	2920.79	2920.79
Omitted Group Mean (Pre-Incentives)		3523.8		3152.53
Total Effect (T × During + Inter.)		0.26, 0.47		541.88, 549.66
p-val Total Effect = 0		0.32, 0.08		0.35, 0.14
Treatment Effect %	12	25, 45	4	19, 19
N Worker-Weeks	935	904	935	904
N Firms	40	40	40	40

Notes: SEs clustered at firm level. All specifications include worker, phase by week, phase by strata fixed effects. Panel A includes bottleneck firms ($\geq$ 1 upstream worker late at baseline); Panel B includes non-bottleneck firms. This regression excludes (a) owners/co-owners who also do stitching work at the firm, and (b) days of firm closure if no production took place at the firm. Output is normalized by product relative to pre-incentive week means. Earnings exclude incentive payments and are coded as 0 for absent workers. Output and earnings are winsorized at the 99th percentile. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Worker Output per hour

	Output per hour | present					
	Daily			Weekly		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	3.97	6.40	4.81	3.04	5.05	5.20
	[5.28]	[5.84]	[6.46]	[5.44]	[6.37]	[6.50]
Treatment × During × Female		-4.24			-3.52	
		[3.54]			[4.30]	
Treatment × During × Late			2.35			0.32
			[5.38]			[6.15]
Treatment × During × Very Late			-6.01			-7.88
			[4.95]			[5.50]
Control Mean (Pre-Incentives)	92.5	92.5	92.5	92.1	92.1	92.1
Omitted Group Mean (Pre-Incentives)		86.67	88.99		84.9	88.08
Total Effect (T × During + Inter.)		2.16	7.16, -1.20		1.52	5.52, -2.68
p-val Total Effect = 0		0.69	0.19, 0.84		0.78	0.39, 0.66
Treatment Effect %	4	2	8, -1	3	2	6, -3
N Worker-Days/Weeks	11771	11771	11676	2398	2398	2378

Notes: All specifications include worker, phase by week, phase by strata fixed effects; SEs (in brackets) clustered at the firm level. *Late* flags workers late at baseline; *Very Late* uses the stricter cutoff. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Treatment Effects on Weekly Firm Output and New Orders

	Firm Output (SD) (worker-reported, daily)		**New Orders (SD)** (owner-reported, weekly)
	All tasks	Last task	
	(1)	(2)	(3)
Panel A: All Firms			
Treatment × During	0.30*	0.21	0.27
	[0.16]	[0.19]	[0.19]
Control Mean (Pre-Incentives)	23419.07	2785.89	4193.61
Control SD (Pre-Incentives)	22826.28	3381.37	3282.05
p-val (T × D)	0.08	0.28	0.16
Treatment Effect %	29	25	21
N Firm-Weeks	525	525	395
N Firms	92	92	87
Panel B: Bottleneck vs Non-Bottleneck Firms			
× Bottleneck	0.46**	0.29	0.29
	[0.21]	[0.29]	[0.28]
× Non-bottleneck	0.13	0.11	0.23
	[0.27]	[0.21]	[0.22]
p-val Bottleneck = Non-bottleneck	0.35	0.62	0.88
N Firm-Weeks	525	525	391
N Firms	92	92	86

Notes: SEs in brackets, clustered at the firm level. Data at the firm-week level; all regressions include firm, week and strata FEs. Firm output counts pieces across all tasks on the production line (column 1) or at the last task, i.e. finished pieces (column 2). New orders are owner-reported material receipts, standardized to pre-incentive product-level means. Panel B interacts Treatment × During with an indicator for bottleneck firms (at least one upstream worker late at baseline). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Treatment Effects on Weekly Firm Revenue, Labor Cost and Profit

		Labor Cost (INR)			**Profit (INR)**		
	Revenue (INR)		(+ incentives)	(+ incentives to late workers)		(− incentives)	(− incentives to late workers)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: All Firms							
Treatment × During	3258	1640	4239*	3105	1617	-981	153
	[4579]	[2301]	[2408]	[2349]	[2712]	[2687]	[2697]
Control Mean (Pre-Incentives)	39275.43	17843.57	17843.57	17843.57	21431.87	21431.87	21431.87
p-val (T × D)	0.48	0.48	0.08	0.19	0.55	0.72	0.95
Treatment Effect %	8	9	24	17	8	-5	1
N Firm-Weeks	525	525	525	525	525	525	525
N Firms	92	92	92	92	92	92	92
Panel B: Bottleneck vs Non-Bottleneck Firms							
× Bottleneck	4127	2360	5299	4188	1767	-1172	-61
	[6297]	[3404]	[3534]	[3449]	[3285]	[3251]	[3265]
× Non-bottleneck	1754	583	2540	1381	1172	-786	374
	[6553]	[2843]	[2982]	[2917]	[4510]	[4476]	[4499]
p-val Bottleneck = Non-bottleneck	0.79	0.69	0.55	0.53	0.92	0.95	0.94
N Firm-Weeks	525	525	525	525	525	525	525
N Firms	92	92	92	92	92	92	92

Notes: SEs in brackets, clustered at the firm level. Data at the firm-week level; all regressions include firm, week and strata FEs. Revenue and labor cost are computed on finished pieces (the last task on the production line). Profit is revenue less labor cost. Columns (3) and (6) add the incentive payments made during the incentive period to labor cost and subtract them from profit; columns (4) and (7) count only the payments to workers late at baseline. Panel B interacts Treatment × During with an indicator for bottleneck firms (at least one upstream worker late at baseline). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A Appendix Tables and Figures

Table A1: Worker Labor Supply and Production Outcomes - Pre Incentives

		Avg. Dev. in Arrival Time before incentives					By Gender		
	All workers	**$\leq$ 0 (Early)**	**$\in$ (0,30] mins (Late)**		**$>$ 30 mins (Very Late)**		**Male**	**Female**	
	mean	mean	mean	**p-value L-E**	mean	**p-value VL-E**	mean	mean	**p-value F-M**
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Attendance, Arrival Times									
Present	0.73	0.78	0.83	0.64	0.75	0.02**	0.72	0.74	0.46
Complied w preferred start time	0.41	0.70	0.44	0.00***	0.08	0.00***	0.54	0.31	0.00***
Dev. in Arrival Time (mins)	7.90	-27.44	14.33	0.00***	54.88	0.00***	-13.51	24.95	0.00***
Dev. in Exit Time (mins)	-37.08	-32.79	-36.34	0.82	-44.55	0.01***	-31.70	-41.35	0.00***
Daily Work Hours	6.64	7.55	7.56	0.03**	6.12	0.00***	6.93	6.41	0.00***
Total # Breaks	1.85	2.10	2.05	0.30	1.75	0.00***	1.89	1.81	0.03**
Total Break Duration - mins	59.39	66.07	67.48	0.53	56.87	0.00***	61.82	57.42	0.04**
Production									
Avg Task Piece Rate (INR)	0.93	1.29	0.85	0.50	0.71	0.93	1.18	0.72	0.01***
Output per hour	91.96	91.07	97.75	0.74	86.66	0.24	92.67	91.40	0.15
Earnings per hour (INR)	74.56	82.90	71.78	0.24	64.89	0.54	84.57	66.60	0.00***
Daily Earnings (INR)	511.50	647.86	558.80	0.05*	397.65	0.00***	609.85	431.83	0.00***
Upstream Task	0.24	0.28	0.26	0.89	0.21	0.07*	0.22	0.25	0.94
Downstream Task	0.23	0.27	0.23	0.09*	0.22	0.02**	0.30	0.17	0.00***
Complex Singer Task	0.08	0.17	0.05	0.07*	0.01	0.00***	0.16	0.02	0.00***
Non-tailoring work	0.11	0.12	0.16	0.14	0.09	0.36	0.11	0.12	0.13
Demographics									
Female	0.55	0.32	0.60	0.00***	0.84	0.00***			
Married	0.87	0.82	0.88	0.05**	0.95	0.00***	0.81	0.93	0.00***
Has children $<$ 14 yrs	0.52	0.52	0.53	0.27	0.49	0.80	0.51	0.52	0.81
Commute $<$ 15 mins	0.84	0.80	0.85	0.33	0.87	0.25	0.81	0.87	0.15
Tenure at Firm	2.85	2.74	3.32	0.72	2.79	0.55	3.12	2.64	0.16
Experience in Industry (yrs)	15.19	16.45	14.52	0.05*	13.69	0.02**	18.31	12.80	0.00***
Owner Baseline Ranking									
Punctual	0.18	0.27	0.17	0.01***	0.08	0.00***	0.22	0.15	0.01**
Reliable	0.19	0.21	0.21	0.47	0.15	0.04**	0.25	0.15	0.00***
Productive	0.18	0.24	0.16	0.02**	0.13	0.00***	0.21	0.16	0.07*
Knowledgable about Prod.	0.18	0.23	0.16	0.02**	0.14	0.13	0.25	0.13	0.00***
benefit most from intervention	0.18	0.23	0.17	0.06*	0.17	0.27	0.21	0.17	0.22
Observations	5084	1980	1399		1349		2277	2807	
Fraction of Workers		0.39	0.30		0.29		0.45	0.55	

Notes: Data at the worker-day level, includes 1-2 weeks of pre-incentives data, excludes days firm was closed. All outcomes coded as 0 if worker was absent. Piece rate and earnings are in INR.

Table A2: Treatment Effects on Deviation from Start Time

	Deviation from Start Time (mins)		
	(1)	(2)	(3)
Treatment × During	-11.18***	-7.62**	0.51
	[2.74]	[2.94]	[2.60]
Treatment × During × Female		-6.17**	
		[2.39]	
Treatment × During × Late			-12.76***
			[2.74]
Treatment × During × Very Late			-27.59***
			[4.49]
Control Mean (Pre-Incentives)	5.59	5.59	5.59
Omitted Group Mean (Pre-Incentives)		-18.31	-30.73
Total Effect (T × During + Inter.)		-13.79	-12.25, -27.08
p-val Total Effect = 0		0.00	0.00, 0.00
Treatment Effect %	-200	-247	-219, -484
Worker, Week FE	X	X	X
Day-of-week FE	X	X	X
N Worker-Days	12437	12437	12308

Notes: All specifications include worker, day-of-week, phase by week, phase by strata fixed effects, from the specification in Equation 1. Standard errors at clustered at the firm level. The sample for these regressions excludes days on which a firm was shut, and no production took place.
Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Treatment Effects on Worker Exit Times

	$\mathbb{I}$**{Left Before}**						**Minutes after**		
	6.30 pm			**Closing Time**			**Regular Closing Time**		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treatment × During	0.00	0.02	0.04	-0.01	-0.01	-0.02	-7.30	-12.54	-14.58
	[0.02]	[0.03]	[0.04]	[0.03]	[0.03]	[0.04]	[7.40]	[8.43]	[9.12]
Treatment × During × Female		-0.03			0.01			9.08*	
		[0.03]			[0.04]			[5.31]	
Treatment × During × Late			-0.02			0.00			7.33
			[0.04]			[0.05]			[7.91]
Treatment × During × Very Late			-0.07**			0.03			17.60*
			[0.04]			[0.06]			[9.51]
Control Mean (Pre-Incentives)	.13	.13	.13	.57	.57	.57	-38.56	-38.56	-38.56
Omitted Group Mean (Pre-Incentives)		.13	.16		.58	.63		-37.67	-38.38
Total Effect (T × During + Inter.)		-0.01	0.01, -0.04		-0.01	-0.01, 0.01		-3.47	-7.24, 3.02
p-val Total Effect = 0		0.74	0.66, 0.22		0.85	0.75, 0.78		0.64	0.32, 0.75
Treatment Effect %	2	-7	11, -30	-2	-1	-2, 3	19	9	19, -8
Worker, Week FE	X	X	X	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X	X	X	X
N Worker-Days	17585	17585	15853	17585	17585	15853	12240	12240	12112

Notes: All specifications include worker, day-of-week, phase by week, phase by strata fixed effects. Standard errors are clustered at the firm level. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).

Table A4: Treatment Effects on # and Duration of Breaks

	# Breaks			Total Break Duration (mins)		
Treatment × During	0.16	0.11	0.06	3.63	2.12	1.62
	[0.10]	[0.11]	[0.14]	[3.16]	[3.47]	[4.32]
Treatment × During × Female		0.09			2.65	
		[0.11]			[3.56]	
Treatment × During × Late			0.07			1.16
			[0.17]			[5.29]
Treatment × During × Very Late			0.30*			6.93
			[0.15]			[5.32]
Control Mean (Pre-Incentives)	1.85	1.85	1.85	59.26	59.26	59.26
Omitted Group Mean (Pre-Incentives)		1.9	2.19		61.83	67.95
Total Effect (T × During + Inter.)		0.20	0.14, 0.36		4.77	2.78, 8.55
p-val Total Effect = 0		0.11	0.30, 0.02		0.20	0.49, 0.08
Treatment Effect %	9	11	7, 19	6	8	5, 14
Worker, Week FE	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X
N Worker-Days	17585	17585	15853	17585	17585	15853

Notes: Data at worker-day level. Outcomes coded 0 if worker was absent. Standard errors at clustered at the firm level. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Treatment Effects on Coordination - Firm Level

	Mins between 1st and last worker	**Mins between the 10th and 90th pctile worker**	**Deviation from start-time: 90th pctile worker**
	(1)	(2)	(3)
T × During	-22.07**	-9.59*	-13.96***
	[10.79]	[5.27]	[4.93]
Control Mean (Pre-Incentives)	128.96	83.93	47.11
Control Mean (Incentive Weeks)	115.32	70.44	31.25
Treatment Effect %	-17	-11	-30
Firm, Week FE	X	X	X
Day-of-week FE	X	X	X
N Firm-Days	2208	2949	2949

Notes: This table reports firm-day level outcomes on coordination in worker arrival times. All specifications include firm, day-of-week, phase by week, phase by strata fixed effects. Standard errors are clustered at the firm level. All outcome variables are in minutes, and are only computed for firm-days where strictly greater than 1 worker was present.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Weekly Worker Earnings

	Weekly Earnings (INR)					
	excl. Incentives			incl. Incentives		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	310.15	164.62	349.08	817.37***	692.90*	944.55**
	[271.34]	[345.07]	[372.16]	[276.26]	[353.76]	[377.99]
Treatment × During × Female		259.67			222.09	
		[261.11]			[268.30]	
Treatment × During × Late			-172.40			-232.74
			[387.26]			[398.11]
Treatment × During × Very Late			52.80			-178.20
			[332.68]			[342.64]
Control Mean (Pre-Incentives)	3505.3	3505.3	3505.3	3505.3	3505.3	3505.3
Omitted Group Mean (Pre-Incentives)		4097.95	4042.68		4097.95	4042.68
Total Effect (T × During + Inter.)		424.28	176.68, 401.88		914.99	711.81, 766.34
p-val Total Effect = 0		0.11	0.60, 0.13		0.00	0.04, 0.01
Treatment Effect %	9	12	5, 11	23	26	20, 22
Worker, Week FE	X	X	X	X	X	X
N Worker-Weeks	2663	2663	2602	2663	2663	2602
N Firms	92	92	92	92	92	92

Notes: SEs clustered at firm level. All specifications include worker, phase by week, phase by strata fixed effects. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Earnings are coded as 0 for workers who are absent on a given day, and are winsorized at the 99th percentile. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Pr. doing non production tasks

	$\mathbb{I}${Did any non-production tasks at the firm}		
	(1)	(2)	(3)
Treatment × During	-0.03	0.01	-0.01
	[0.03]	[0.03]	[0.04]
Treatment × During × Female		-0.06**	
		[0.03]	
Treatment × During × Late			-0.04
			[0.03]
Treatment × During × Very Late			-0.01
			[0.04]
Control Mean (Pre-Incentives)	.16	.16	.16
Omitted Group Mean (Pre-Incentives)		.16	.16
Total Effect (T × During + Inter.)		-0.05	-0.05, -0.02
p-val Total Effect = 0		0.14	0.15, 0.64
Treatment Effect %	-16	-33	-32, -12
Worker, Week FE	X	X	X
Day-of-week FE	X	X	X
N Worker-Days	12425	12425	12314

Notes: SEs clustered at firm level. All specifications include worker, day-of-week, phase by week, phase by strata fixed effects. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A8: Treatment Effects on Helper Attendance - Firm Level

	Fraction Helpers Present
	(1)
Treatment × During Incentives	0.03
	[0.04]
Control Mean (Pre-Incentives)	.83
Control Mean (Incentive Weeks)	.81
Treatment Effect %	4
Firm, Week FE	X
Day-of-week FE	X
N Firm-Days	3039

Notes: This table reports firm-day level outcomes on helper attendance. All specifications include firm, day-of-week, phase by week, phase by strata fixed effects. Standard errors are clustered at the firm level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Baseline Firm Characteristics by Bottleneck Status

	Non-Bottleneck	Bottleneck	B - NB	*p-value*
	Mean/SD	Mean/SD	Difference	
Firm Characteristics				
# Tailors	5.70	6.88	1.79	0.01
	(2.71)	(3.05)		
# Helpers	1.15	1.73	0.81	0.00
	(1.10)	(1.17)		
Fraction female workers	0.53	0.64	0.08	0.16
	(0.27)	(0.24)		
# late workers in firm	1.57	4.17	2.82	0.00
	(1.43)	(2.12)		
Avg. active orders per day	1.95	2.24	0.48	0.01
	(0.61)	(0.88)		
Total # Machines	10.32	12.23	2.82	0.00
	(3.97)	(4.42)		
Owner directly supervises workers	0.90	0.98	0.07	0.24
	(0.30)	(0.14)		
Firm Age (yrs)	6.53	6.98	0.47	0.71
	(6.69)	(5.12)		
Export orders	0.10	0.19	0.10	0.23
	(0.30)	(0.40)		
Only stitching section	0.55	0.63	-0.02	0.86
	(0.50)	(0.49)		
Weekly Output	3912.12	4545.83	897.22	0.16
	(2116.57)	(2886.94)		
Business Finances				
Revenues (last 30 days)	98783.78	156946.00	77163.20	0.00
	(73445.42)	(126551.47)		
Profits (last 30 days)	26270.27	33548.16	12587.49	0.24
	(16517.08)	(54894.03)		
Total Costs - wage bill (last 30 days)	28056.00	36690.35	12413.56	0.01
	(20092.20)	(18849.50)		
Owner Demographics				
Age	39.80	38.21	-1.10	0.59
	(8.87)	(8.16)		
Male	0.90	0.77	-0.11	0.18
	(0.30)	(0.43)		
Completed Secondary School	0.60	0.58	-0.02	0.87
	(0.50)	(0.50)		
Commute Time (mins)	7.70	7.25	-0.49	0.77
	(7.80)	(8.46)		
Experience in Industry (yrs)	20.62	17.62	-2.48	0.30
	(10.67)	(9.43)		
Owner does stiching work	0.85	0.88	0.02	0.77
	(0.36)	(0.32)		
Owner stitches as a regular tailor	0.42	0.44	-0.05	0.59
	(0.50)	(0.50)		
Observations	40	52		

Notes: This table reports means, mean differences by firm bottleneck status and p-values for baseline firm and owner characteristics. A firm is a bottleneck firm if at least one upstream worker arrived late at baseline. Mean differences and p-values are estimated using a regression of each variable on a bottleneck indicator with strata fixed effects, with robust standard errors.

Table A10: Baseline Worker Characteristics by Bottleneck Status

	Non-Bottleneck	Bottleneck	B - NB	*p-value*
	Mean/SD	Mean/SD	Difference	
Age	37.25	36.74	-1.11	0.30
	(9.06)	(8.78)		
Male	0.50	0.40	-0.09	0.11
	(0.50)	(0.49)		
Married	0.84	0.87	0.05	0.22
	(0.37)	(0.34)		
Has children $<$ 14 yrs	0.84	0.90	-0.03	0.74
	(0.96)	(1.35)		
Has elderly member in HH	0.39	0.38	-0.02	0.71
	(0.49)	(0.49)		
Related to owner	0.16	0.24	0.04	0.33
	(0.37)	(0.43)		
Native to TN	0.92	0.95	0.03	0.31
	(0.27)	(0.22)		
Commute $<$ 15 mins	0.84	0.84	-0.01	0.79
	(0.37)	(0.37)		
Completed Secondary School	0.40	0.46	0.06	0.21
	(0.49)	(0.50)		
Experience in Industry (yrs)	15.51	14.44	-2.38	0.03
	(9.80)	(9.24)		
Tenure at Firm (yrs)	2.39	2.53	0.06	0.90
	(3.78)	(3.21)		
Observations	223	368		

Notes: This table reports means, mean differences by firm bottleneck status and p-values for baseline worker characteristics. Mean differences and p-values are estimated using a regression of each variable on a bottleneck indicator with strata fixed effects, with standard errors clustered at the firm level.

Table A11: Treatment Effects on Weekly Firm Output and New Orders, Weeks 1–3

	Firm Output (SD) (worker-reported, daily)			**New Orders (SD)** (owner-reported, weekly)
	All tasks	Last task	All but last	
	(1)	(2)	(3)	(4)
Panel A: All Firms				
Treatment × During	0.33*	0.31	0.33*	0.26
	[0.17]	[0.20]	[0.18]	[0.19]
Control Mean (Pre-Incentives)	23419.07	2785.89	20633.18	4193.61
Control SD (Pre-Incentives)	22826.28	3381.37	20157.05	3282.05
p-val (T × D)	0.06	0.11	0.07	0.17
Treatment Effect %	32	38	32	20
N Firm-Weeks	434	434	434	334
N Firms	92	92	92	87
Panel B: Bottleneck vs Non-Bottleneck Firms				
× Bottleneck	0.46*	0.42	0.50**	0.30
	[0.24]	[0.30]	[0.24]	[0.27]
× Non-bottleneck	0.20	0.15	0.17	0.18
	[0.26]	[0.23]	[0.28]	[0.22]
p-val Bottleneck = Non-bottleneck	0.48	0.49	0.38	0.72
N Firm-Weeks	434	434	434	331
N Firms	92	92	92	86

Notes: SEs in brackets, clustered at the firm level. Data at the firm-week level; all regressions include firm, week and strata FEs. Firm output counts pieces across all tasks on the production line (column 1) or at the last task, i.e. finished pieces (column 2). Column (3) counts pieces at all tasks except the last. New orders are owner-reported material receipts, standardized to pre-incentive product-level means. Panel B interacts Treatment × During with an indicator for bottleneck firms (at least one upstream worker late at baseline). Restricted to weeks 1–3 of the incentive period. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A12: Flexibility by Firm Size

	Small Firms		Large Firms	
	Mean	SD	Mean	SD
Pay Structure				
Piece Rate	0.90	(0.30)	0.60	(0.49)
Shift Rate	0.06	(0.23)	0.40	(0.49)
Average Monthly Earnings (Rs.)	16087.43	(5899.93)	16713.75	(5323.63)
Work Timings				
Timings Same	0.49	(0.50)	0.77	(0.42)
Timings Strict	0.15	(0.36)	0.50	(0.51)
Entry time is recorded	0.03	(0.17)	0.50	(0.51)
Pay cut if late	0.06	(0.23)	0.40	(0.49)
Margin of Lateness				
Acceptable to be 5 minutes late	0.97	(0.17)	0.94	(0.24)
Acceptable to be <= 30 minutes late	0.91	(0.29)	0.69	(0.47)
Acceptable to be > 30 minutes late	0.85	(0.36)	0.62	(0.49)
Observations	70		48	

Notes: SD in parenthesis (even columns). Small firms are defined as firms with up to 20 workers, and where the owner is often both a tailor (either full or part-time) and manager of the firm. Large firms include firms with at least 50 workers, where ownership and factory-floor management are separate.

Figure A1: Self-Reported Idle Time among Workers

